Detection of Man-in-the-Middle Attacks in Software-Defined Networks Using Flow-Level Monitoring

Progress presentation aligned with the grant theme on intelligent network design, deployment, and security in Nepal

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Problem Statement

Local networks are inherently vulnerable to silent interception

ARP the protocol that maps IP addresses to MAC addresses - has no built-in authentication.
Any device can claim any IP, poisoning neighbors' caches and silently redirecting their traffic.

Attackers exploit this to become an invisible Man-in-the-Middle

By continuously sending fake ARP replies, an attacker intercepts all traffic between victim and server reading credentials, injecting content, and forwarding packets so neither party notices.

Rule-based detection alone cannot catch subtle or low-rate attacks

Static ARP inspection flags obvious mis matches but fails against patient attackers who throttle their poisoning rate or mimic legitimate traffic timing - producing dangerous false negatives.

Flow-level behavioral patterns expose what packet-level rules miss

A relay-based MITM creates measurable statistical anomalies as asymmetric byte counts, extra latency jitter, irregular packet-size distributions that ML models can learn to recognise.

A hybrid two-layer system is needed -and directly supports LISPAN goals

Combining SDN-based ARP detection with a CNN+BILSTM+Attention model provides fast, accurate, and adaptive MITM detection, deployable for Nepal's emerging SDN, loT, and Telco networks.

Research Objectives

Build Real-Time MITM Detection System

Implement a **dual-layer defense:** rule-based inspection via Ryu SDN controller (Layer 1) for instant blocking, and ML-based flow analysis (Layer 2) for subtle attack patterns.

Train & Evaluate CNN + BiLSTM + Attention Model

Train a hybrid deep learning model on ~148,686 real-world labeled network flows from 5 datasets (CIC, UQ, IoT, GitHub). Achieve publication-grade metrics via **5-fold cross-validation.**

Validate in Mininet SDN Simulation Environment

Simulate real ARP poisoning attacks using Mininet virtual network (victim, server, attacker hosts) with Ryu SDN controller in Docker - demonstrating **full detection and mitigation lifecycle**.

Contribute to Nepal's Network Security Research

Align with LISPAN grant goals: provide ML-based anomaly detection, SDN security orchestration, and **deployable security framework** applicable to Nepali ISPs and IoT networks.

Methods / Research Plan

Planned workflow for the study and system implementation

1. SDN testbed

Mininet + OVS + Ryu
Victim, server, attacker
ARP spoofing scenarios

2. Data pipeline

Merge labelled datasets
Preprocess and binarise labels
Create unified feature table

3. Feature learning

Engineer asymmetry and ratio features
Select top 25 features with RF-RFE

4. Detection model

Train CNN-BiLSTM-Attention
Validate with holdout test and 5-fold CV

5. Controller integration

Load scaler + features + model
Monitor flows in real time
Trigger alerts / blocking

Progress Till Date - Implementation

Completed technical work so far

Completed pipeline

- Literature review on SDN security, MITM attacks, ARP spoofing, and flow-level monitoring completed.
- SDN testbed built using Mininet, Open vSwitch, and Ryu controller.
- Normal and malicious traffic scenarios prepared for flow collection and controller testing.
- Five heterogeneous labelled datasets merged and preprocessed into a unified dataset.

Model and deployment readiness

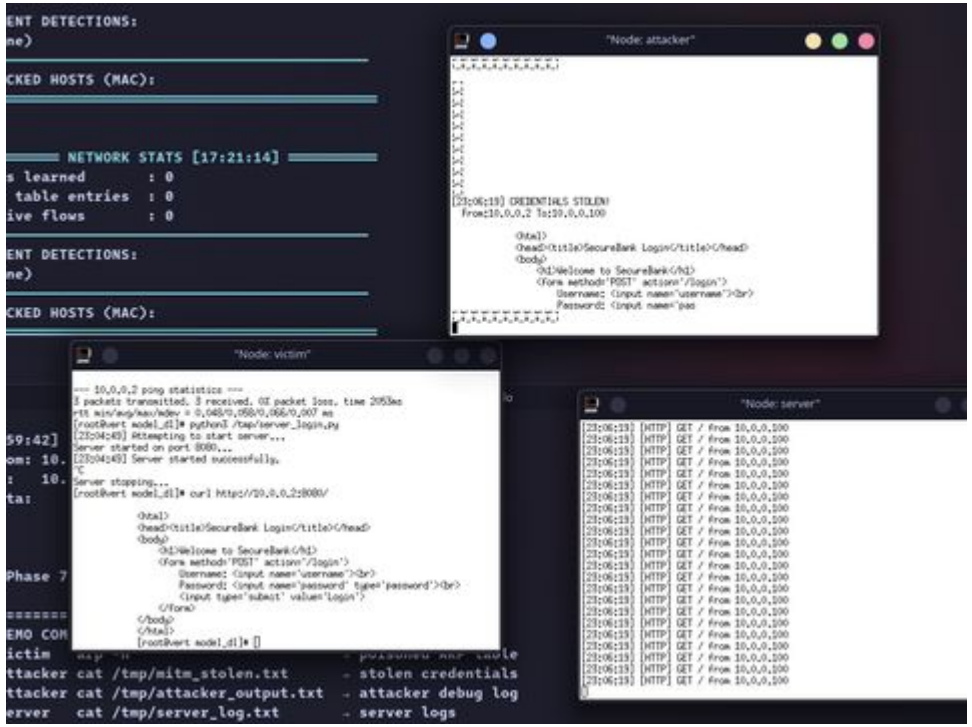
- Domain-specific engineered features created for MITM behaviour analysis.
- Top 25 discriminative features selected using RF-RFE.
- CNN-BiLSTM-Attention model trained and saved together with scaler and feature files.
- Docker-based controller deployment workflow prepared for reproducibility.

Progress Till Date - Results

```
[21:22:56] [HTTP ] 10.0.0.12:55555 → 10.0.0.2:8080
[21:22:57] [HTTP ] 10.0.0.12:55555 → 10.0.0.2:8080
[21:22:57] [HTTP ] 10.0.0.12:55555 → 10.0.0.2:8080

🚨 MITM ATTACK DETECTED — SESSION HIJACKING
Time:      21:22:57
Host IP:   10.0.0.12
MAC:       00:00:00:00:00:12
Details:   RST injection (rst_ratio=0.33, acks=10, pkts=15)
Action:    MAC+IP blocked — DROP rule installed

[21:23:04] [ARP  ] REQUEST  10.0.0.100 (00:00:00:00:00:03) -> 10.0.0.1 | switch=1 port=3
[21:23:05] [ARP  ] REQUEST  10.0.0.100 (00:00:00:00:00:03) -> 10.0.0.2 | switch=1 port=3
[21:23:05] [ARP  ] REPLY    10.0.0.1 (00:00:00:00:00:01) -> 10.0.0.100 | switch=1 port=1
[21:23:05] [ARP  ] REPLY    10.0.0.2 (00:00:00:00:00:02) -> 10.0.0.100 | switch=1 port=2
[21:23:05] [ARP  ] REPLY    10.0.0.2 (00:00:00:00:00:03) -> 10.0.0.1 | switch=1 port=3

🚨 MITM ATTACK DETECTED — ARP POISONING
Time:      21:23:05
Host IP:   10.0.0.2
MAC:       00:00:00:00:00:03
Details:   IP 10.0.0.2 known-MAC=00:00:00:00:00:02 fake-MAC=00:00:00:00:00:03
Action:    MAC+IP blocked — DROP rule installed
```

```
═══ NETWORK STATS [04:22:38] ═══
Switches connected : 3
MACs learned       : 24
ARP table entries  : 9
Active flows       : 213

FLOW TABLE:
Flow                      Pkts    Bytes  Dur      S
10.0.5.100 → 10.0.1.10      10     1167  78s
10.0.5.100 → 10.0.1.10      11     1219  77s
10.0.5.100 → 10.0.1.10      15     1851  77s
10.0.5.101 → 10.0.1.10       7      420  77s
10.0.5.100 → 10.0.1.10       7      420  77s

DETECTIONS:
 [04:20:12] SESSION HIJACKING      10.0.2.52 blocke
 [04:20:16] ARP POISONING          10.0.2.50 blocke
 [04:21:21] SESSION HIJACKING      10.0.5.100 block

BLOCKED:
 MACs: 7a:90:d2:43:47:d2, 00:00:00:00:02:52, 00:00:
 IPs : 10.0.2.52, 10.0.2.50, 10.0.5.100, 10.0.2.51
```

```
LOCKED:
MACs: (None)
IPs : (None)

21:24:11] [HTTP ] 10.0.2.51:44300 → 10.0.1.10:443

🚨 MITM ATTACK DETECTED — SSL STRIPPING
Time:      21:24:11
Host IP:   10.0.2.51
MAC:       00:00:00:00:02:51
Details:   TLS port flow detected (port=443, pkts=5)
Action:    MAC+IP blocked — DROP rule installed
```



Association with Grant and National Contribution

How the project contributes to Nepal

Association with LISPAN grant title

- Directly fits the grant theme: intelligent network design, deployment, and security prospects analysis in Nepal.
- Targets the SDN security dimension of the grant through AI/ML-based anomaly detection and SDN-enabled response.
- Provides a focused, testbed-driven case study under the broader next-generation network research agenda.

National contribution

- Improves the security readiness of future ISP, campus, enterprise, and government SDN deployments in Nepal.
- Supports national digital transformation through secure, scalable, and efficient communication infrastructure.
- Builds local research capacity using a practical and low-cost cybersecurity testbed relevant to Nepal.